RESEARCH MAP · 2026

可控语音合成：从 Model 到 Agent

Controllable Speech Synthesis: From Generative Models to Interactive Voice Agents

范围：可控 TTS、上下文自适应语音生成、Agentic Speech Production、Interactive Voice Agent；重点优先整理具有规划、检索、工具、批评反馈、流式状态或实时行动机制的工作。

更新时间：2026-08-15。论文按首次公开时间排序；同一工作若同时贡献数据集、训练方法或评测，可在对应证据层交叉列出。

核心判断

- **控制对象发生变化：**从音高、韵律、音色和情绪，扩展到角色、对话行为、话轮时机、工具调用与任务行动。
- **控制来源发生变化：**从用户显式 Prompt，转向系统基于语音、上下文、Persona、目标和反馈自主推断。
- **Agent 的分界不在名称：**只有当规划、检索、批评或在线状态决策真正影响语音生成/交互时，才进入 Agent 主线。
- Agentic 过渡早于 2026：AudioGPT（2023）开启 LLM–音频工具编排，SpeechAgents 与 WavCraft（2024）进一步引入多智能体沟通模拟和可组合音频工作流。
- **2026 年的关键跃迁：**AgentSteerTTS、Harness TTS、ActorMind、PRISM、DuplexSLA 等开始把可控语音纳入闭环决策。

工作分类与判断边界

Level	Control decision	Agent mechanism	Role of speech	Inclusion boundary
M · Model	User/reference explicitly supplies emotion, prosody, timbre, voice or style.	None required; model executes a condition.	Generated artifact	Controllable TTS anchor; not an agent.
C · Context-aware	System infers how to speak from user speech, history, affect, persona or goal.	Context inference; optional retrieval or emotion reasoning.	Context-aligned response	Agent-adjacent unless an explicit decision loop exists.
A · Agentic Speech	Planner chooses style/content/voice and may revise it.	Planning, retrieval/tool routing, critic, feedback or multi-agent orchestration.	Planned/revised speech artifact	Agent loop acts on speech generation, often offline.
V · Voice Agent	System decides how and when to speak while interacting online.	Streaming state, reasoning, tool/action, turn policy and feedback.	Real-time action	Speech participates in a live reason-plan-act loop.

符号说明：M = Model-level；C = Context-aware；A = Agentic Speech；V = Interactive Voice Agent。该四级划分是本调研使用的工作分类，不宜称为社区统一标准。

1. Surveys and Positioning

直接综述用于界定可控属性、模型架构和评测；邻近综述用于补足从语音模型到全双工交互的系统脉络。

1. **"Towards Controllable Speech Synthesis in the Era of Large Language Models: A Survey".** *Tianxin Xie et al.* EMNLP 2025 / arXiv 2024. [Paper]
2. **"WavChat: A Survey of Spoken Dialogue Models".** *Shengpeng Ji et al.* arXiv 2024. [Paper]
3. **"From Turn-Taking to Synchronous Dialogue: A Survey of Full-Duplex Spoken Language Models".** *Yuxuan Chen and Haoyuan Yu* arXiv 2025. [Paper]
4. **"A Survey of Full-Duplex Spoken Dialogue Systems: Architectural Hierarchy, Interaction Ontology, and Decision State Machine".** *Jingyu Lu et al.* arXiv 2026. [Paper]

2. Model-Level Controllable TTS（模型级可控合成）

这一层回答“给定控制条件，模型能否稳定地产生目标语音”。它构成 Agent 的执行器基础，但控制决策仍主要来自用户、参考语音或预先定义的 Prompt。

1. **"Style Tokens: Unsupervised Style Modeling, Control and Transfer in End-to-End Speech Synthesis".** *Yuxuan Wang et al.* arXiv 2018. [Paper]
2. **"Mellotron: Multispeaker Expressive Voice Synthesis by Conditioning on Rhythm, Pitch and Global Style Tokens".** *Rafael Valle et al.* arXiv 2019. [Paper]
3. **"PromptTTS: Controllable Text-to-Speech with Text Descriptions".** *Zhifang Guo et al.* arXiv 2022. [Paper]
4. **"InstructTTS: Modelling Expressive TTS in Discrete Latent Space with Natural Language Style Prompt".** *Dongchao Yang et al.* arXiv 2023. [Paper]
5. **"NaturalSpeech 2: Latent Diffusion Models are Natural and Zero-Shot Speech and Singing Synthesizers".** *Kai Shen et al.* arXiv 2023. [Paper]
6. **"PromptTTS 2: Describing and Generating Voices with Text Prompt".** *Yichong Leng et al.* ICLR 2024 / arXiv 2023. [Paper]
7. **"Natural Language Guidance of High-Fidelity Text-to-Speech with Synthetic Annotations".** *Dan Lyth and Simon King* arXiv 2024. [Paper]
8. **"NaturalSpeech 3: Zero-Shot Speech Synthesis with Factorized Codec and Diffusion Models".** *Zeqian Ju et al.* arXiv 2024. [Paper]
9. **"ControlSpeech: Towards Simultaneous Zero-shot Speaker Cloning and Zero-shot Language Style Control With Decoupled Codec".** *Shengpeng Ji et al.* arXiv 2024. [Paper]
10. **"CosyVoice: A Scalable Multilingual Zero-shot Text-to-speech Synthesizer based on Supervised Semantic Tokens".** *Zhihao Du et al.* arXiv 2024. [Paper]
11. **"CosyVoice 3: Towards In-the-wild Speech Generation via Scaling-up and Post-training".** *Zhihao Du et al.* arXiv 2025. [Paper]
12. **"Qwen3-TTS Technical Report".** *Qwen Team et al.* arXiv 2026. [Paper]

3. Context-Aware Conversational Speech Synthesis（上下文自适应）

这一层开始回答“在当前对话中应该怎么说”。系统利用用户语音、历史、情绪、Persona 或检索到的相似对话推断表达方式，但并非每项工作都具有完整 Agent 闭环。

1. **"Talk With Human-like Agents: Empathetic Dialogue Through Perceptible Acoustic Reception and Reaction".** *Haoqiu Yan et al.* arXiv 2024. [Paper]
2. **"Style-Talker: Finetuning Audio Language Model and Style-Based Text-to-Speech Model for Fast Spoken Dialogue Generation".** *Yinghao Aaron Li et al.* arXiv 2024. [Paper]
3. **"JELLY: Joint Emotion Recognition and Context Reasoning with LLMs for Conversational Speech Synthesis".** *Jun-Hyeok Cha et al.* arXiv 2025. [Paper]
4. **"Retrieval-Augmented Dialogue Knowledge Aggregation for Expressive Conversational Speech Synthesis".** *Rui Liu et al.* arXiv 2025. [Paper]
5. **"Chain-Talker: Chain Understanding and Rendering for Empathetic Conversational Speech Synthesis".** *Yifan Hu et al.* arXiv 2025. [Paper]
6. **"Voicing Personas: Rewriting Persona Descriptions into Style Prompts for Controllable Text-to-Speech".** *Yejin Lee et al.* arXiv 2025. [Paper]
7. **"EmoNews: A Spoken Dialogue System for Expressive News Conversations".** *Ryuki Matsuura et al.* arXiv 2025. [Paper]

8. **"ES4R: Speech Encoding Based on Prepositive Affective Modeling for Empathetic Response Generation".** *Zhuoyue Gao et al.* arXiv 2026. [Paper]
9. **"Sympatheia: Emotionally Adaptive Voice Assistant with Continuous Affect Conditioning".** *Sukru Samet Dindar et al.* arXiv 2026. [Paper]
10. **"Self-EmoQ: Plutchik-Guided Value-based Planning to Drive Streaming Emotional TTS".** *Yue Zhao et al.* arXiv 2026. [Paper]

11. **"Borderless Long Speech Synthesis".** *Xingchen Song et al.* arXiv 2026. [Paper]
12. **"AuEmoChat: Authentic Emotion Understanding and Rendering for Conversational Speech Synthesis".** *Zhenqi Jia et al.* ACM MM 2026 / arXiv 2026. [Paper]

4. Agentic Speech Generation and Orchestration（重点）

本节优先收录存在显式 Planner、Retriever、Critic、Feedback、Tool Orchestration 或 Multi-Agent 协作，并且这些机制真实控制语音生成、角色配音、语音修复或长音频编排的工作。条目同时标注语音是否为核心产物，避免把仅使用 TTS 模块的通用 Agent 误判为可控语音方法。

1. **"AudioGPT: Understanding and Generating Speech, Music, Sound, and Talking Head".** *Rongjie Huang et al.* arXiv 2023. [Paper]

定位：早期 LLM 音频工具编排锚点；ChatGPT 负责意图解析与模型选择，语音生成是多种音频能力之一，并非专门的闭环 TTS。

2. **"SpeechAgents: Human-Communication Simulation with Multi-Modal Multi-Agent Systems".** *Dong Zhang et al.* arXiv 2024. [Paper]

定位：以多模态多智能体模拟人类沟通，角色状态和交互历史影响文本、语音及非语言行为；属于多 Agent 语音交互生产。

3. **"WavCraft: Audio Editing and Generation with Large Language Models".** *Jinhua Liang et al.* arXiv 2024. [Paper]

定位：LLM 将自然语言请求拆解为可执行音频工具链，支持语音/音频生成与编辑；是 Agentic Audio Workflow，而非单一 TTS 模型。

4. **"PodAgent: A Comprehensive Framework for Podcast Generation".** *Yujia Xiao et al.* arXiv 2025. [Paper]

定位：Host–Guest–Writer 多智能体协作，联合内容规划、角色配音和表达性语音生成；属于离线 Agentic Speech Production。

5. **"DialogueAgents: A Hybrid Agent-Based Speech Synthesis Framework for Multi-Party Dialogue".** *Xiang Li et al.* ICME 2025 / arXiv 2025. [Paper]

定位：Script Writer、Speech Synthesizer 与 Dialogue Critic 构成迭代闭环，并产出 MultiTalk；Agent 作用于脚本与语音质量控制。

6. **"A Multi-Agent Framework for Automated Qinqiang Opera Script Generation Using Large Language Models".** *Gengxian Cao et al.* arXiv 2025. [Paper]

定位：多 Agent 协作完成戏曲脚本、视觉与 TTS 制作；语音是跨模态生产链的执行环节，列为邻近的 Agentic Speech Production。

7. **"A Multi-Agent AI Framework for Immersive Audiobook Production through Spatial Audio and Neural Narration".** *Shaja Arul Selvamani and Nia D'Souza Ganapathy* arXiv 2025. [Paper]

定位：把文本理解、角色/旁白规划、神经 TTS 与空间音频编排组合为多 Agent 有声书生产流程。

8. **"AI4Reading: Chinese Audiobook Interpretation System Based on Multi-Agent Collaboration".** *Minjiang Huang et al.* ACL 2025 Demo / arXiv 2025. [Paper]

定位：11 个主题分析、案例分析、编辑、叙述与校对 Agent 共同规划脚本并合成播客式有声解读，强调内容正确性和可理解性。

9. **"STEAMROLLER: A Multi-Agent System for Inclusive Automatic Speech Recognition for People who Stutter".** *Ziqi Xu et al.* arXiv 2026. [Paper]

定位：ASR、多 Agent 语义修复与语音合成形成实时无障碍 speech-to-speech 流程；控制目标是语义保持与流利化，而非风格生成。

10. **"ActorMind: Emulating Human Actor Reasoning for Speech Role-Playing".** *Xi Chen et al.* arXiv 2026. [Paper]

定位：Eye–Ear–Brain–Mouth 多 Agent 推理链把角色、场景和情绪状态转为个性化语音表演。

11. **"AuDirector: A Self-Reflective Closed-Loop Framework for Immersive Audio Storytelling".** *Yiming Ren et al.* arXiv 2026. [Paper]

定位：角色规划、语音检索、合成审查、自我修正与人类反馈共同形成长音频制作闭环。

12. **"AgentSteerTTS: A Multi-Agent Closed-Loop Framework for Composite-Instruction Text-to-Speech".** *Bin Kang et al.* arXiv 2026. [Paper]

定位：Retrieval Agent、Synthesis Agent 与 Fast–Slow Feedback Agent 共同解决复合指令的语义–声学对齐。

13. **"Audio-Oscar: A Multi-Agent System for Complex Audio Scene Generation, Orchestration, and Refinement".** *Yifan Duan et al.* arXiv 2026. [Paper]

定位：面向复杂音频场景的规划、角色/声线设计、语音与非语音生成、时间线编排和反馈式精修。

14. **"PRISM: Prosody-Integrated Multi-Agent Reasoning Framework for Empathetic Spoken Dialogue".** *Wen Zhang et al.* arXiv 2026. [Paper]

定位：将韵律感知、响应生成和语音合成解耦为协作 Agent，并按需调用外部知识工具。

15. **"Harness TTS: Towards Context-Aware Expressive Speech Synthesis with Harness Layer".** *Shengfan Shen et al.* arXiv 2026. [Paper]
 定位：LLM Planner 根据显式、隐式和冲突上下文路由风格 Prompt Tool，提供可审计、低延迟的 TTS 控制层。

5. Interactive Voice Agents（重点）

这一层把语音从“生成结果”提升为“实时行动”：系统需要持续监听、决定何时说、是否打断或 Backchannel，并将推理、角色、工具或结构化 Action 与语音同步。

1. **"Cloning a Conversational Voice AI Agent from Call Recording Datasets for Telesales".** *Krittanon Kaewtawee et al.* arXiv 2025. [Paper]

 定位：从优秀人工坐席通话中抽取知识、Playbook 与 Prompt，串联 ASR–LLM–TTS 流式管线；控制重点是领域话术与任务行为。

2. **"SpeechAgent: An End-to-End Mobile Infrastructure for Speech Impairment Assistance".** *Haowei Lou et al.* arXiv 2025. [Paper]

 定位：LLM 推理与语音处理模块按障碍类型提供自适应支持，并在移动/边缘设备上实时生成可交流语音。

3. **"F-Actor: Controllable Conversational Behaviour in Full-Duplex Models".** *Maike Züfle et al.* arXiv 2026. [Paper]
 定位：把声线、话题、Backchannel、打断和发起行为统一为可指令控制的全双工对话行为。
4. **"LTS-VoiceAgent: A Listen-Think-Speak Framework for Efficient Streaming Voice Interaction via Semantic Triggering and Incremental Reasoning".** *Wenhao Zou et al.* arXiv 2026. [Paper]
 定位：语义触发器与 Thinker/Speaker 双角色编排，使推理在用户尚未说完时增量展开。
5. **"PersonaPlex: Voice and Role Control for Full Duplex Conversational Speech Models".** *Rajarshi Roy et al.* arXiv 2026. [Paper]
 定位：将角色 Prompt 与声线克隆注入全双工模型；实时角色控制突出，但工具与长程规划仍有限。
6. **"DuplexSLA: A Full-Duplex Spoken Language Model with Synchronized Speech, Language, and Action".** *Haoyang Zhang et al.* arXiv 2026. [Paper]
 定位：在同一 160 ms 时钟上联合解码用户音频、助手语音和结构化 Action，实现边说边规划与工具调用。
7. **"Adaptive Turn-Taking for Real-time Multi-Party Voice Agents".** *Soumyajit Mitra et al.* Interspeech 2026 / arXiv 2026. [Paper]
 定位：ModeratorLM 根据分配角色和多人上下文自主决定是否接管话轮，并提供 CoT 推理变体。
8. **"VoxMind: An End-to-End Agentic Spoken Dialogue System".** *Tianle Liang et al.* ACL 2026. [Paper]
 定位：端到端语音智能体主线，将语音交互、推理和任务执行作为统一系统评估。
9. **"JoyAI-Talker: Full-Duplex Speech Interactive Large Model Built for Empathetic Voice Agents".** *Yinhao Bai et al.* arXiv 2026. [Paper]
 定位：Thinker–Talker、Persona-Adaptive Empathetic Response 与 Joy-Duplex 共同控制内容、声学表达和实时对话状态。

6. Supporting Full-Duplex / Omni Foundations

以下工作提供实时、流式或全双工基础能力，但仅凭该能力不能自动归为 Agent；它们主要构成 Voice Agent 的系统底座。

1. **"Beyond Turn-Based Interfaces: Synchronous LLMs as Full-Duplex Dialogue Agents".** *Bandhav Veluri et al.* EMNLP 2024. [Paper]
2. **"Moshi: A Speech-Text Foundation Model for Real-Time Dialogue".** *Alexandre Défossez et al.* arXiv 2024. [Paper]
3. **"Qwen2.5-Omni Technical Report".** *Qwen Team et al.* arXiv 2025. [Paper]
4. **"ChipChat: Low-Latency Cascaded Conversational Agent in MLX".** *Tatiana Likhomanenko et al.* ASRU 2025 / arXiv 2025. [Paper]
5. **"MiniCPM-o 4.5: Towards Real-Time Full-Duplex Omni-Modal Interaction".** *Junbo Cui et al.* arXiv 2026. [Paper]

7. Training, Alignment and Reward

从 Model 到 Agent 的另一条主线是由一次性监督转向偏好、奖励、用户交互反馈和多维交互行为优化。

1. **"SpeechAlign: Aligning Speech Generation to Human Preferences".** *Dong Zhang et al.* arXiv 2024. [Paper]
2. **"WavReward: Spoken Dialogue Models With Generalist Reward Evaluators".** *Shengpeng Ji et al.* arXiv 2025. [Paper]
3. **"Aligning Spoken Dialogue Models from User Interactions".** *Anne Wu et al.* arXiv 2025. [Paper]
4. **"Multi-Faceted Interactivity Alignment in Full-Duplex Speech Models".** *Atsumoto Ohashi et al.* arXiv 2026. [Paper]

8. Datasets and Evaluation Benchmarks

评测也从 MOS、相似度和 Prompt Following，扩展到上下文一致性、角色遵循、工具正确性、任务完成、打断鲁棒性和实时交互。

1. **"DailyTalk: Spoken Dialogue Dataset for Conversational Text-to-Speech".** *Keon Lee et al.* arXiv 2022. [Paper]
2. **"SpeechCraft: A Fine-grained Expressive Speech Dataset with Natural Language Description".** *Zeyu Jin et al.* arXiv 2024. [Paper]
3. **"DialogueAgents: A Hybrid Agent-Based Speech Synthesis Framework for Multi-Party Dialogue (MultiTalk)".** *Xiang Li et al.* ICME 2025. [Paper]
4. **"VoiceAgentBench: Are Voice Assistants Ready for Agentic Tasks?".** *Dhruv Jain et al.* arXiv 2025. [Paper]
5. **"ActorMind: Emulating Human Actor Reasoning for Speech Role-Playing (ActorMindBench)".** *Xi Chen et al.* arXiv 2026. [Paper]
6. **"Adaptive Turn-Taking for Real-time Multi-Party Voice Agents (RolePlayConv)".** *Soumyajit Mitra et al.* Interspeech 2026. [Paper]
7. **"Audio-Oscar: A Multi-Agent System for Complex Audio Scene Generation, Orchestration, and Refinement (ASG-Bench)".** *Yifan Duan et al.* arXiv 2026. [Paper]
8. **"τ-Voice: Benchmarking Full-Duplex Voice Agents on Real-World Domains".** *Soham Ray et al.* arXiv 2026. [Paper]
9. **"Full-Duplex-Bench-v3: Benchmarking Tool Use for Full-Duplex Voice Agents Under Real-World Disfluency".** *Guan-Ting Lin et al.* arXiv 2026. [Paper]
10. **"ISCSLP 2026 CoT-TTS Challenge: Chain-of-Thought Reasoning for Context-Aware Text-to-Speech".** *Wei Xue et al.* arXiv 2026. [Paper]
11. **"RW-Voice-EQ Bench: A Real World Benchmark for Evaluating Voice AI Systems".** *David Ayllon et al.* arXiv 2026. [Paper]
12. **"DuplexWorld: Can Voice Agents Help You Get Through the Day?".** *Aryan Vijay Bhosale et al.* arXiv 2026. [Paper]

9. Representative Model-to-Agent Comparison

Table 1 compares representative works along decision source and agent mechanisms. ✓ = native and explicitly evaluated; △ = partial, external, or not the primary contribution; − = not central. Newly added early orchestration, production-agent and domain-agent works make the Model-to-Agent transition visible before the 2026 closed-loop TTS wave.

Method	Level	Control source	Context	Plan / Reason	Retrieval / Tool	Feedback / Critic	Emotion / Style	Persona / Voice	Real-time / Duplex	Speech as Action	Release
GST-Tacotron	M	Reference / latent token	-	-	-	-	Style, speed	Voice/style	-	-	2018.03
PromptTTS	M	Natural-language prompt	-	-	-	-	Style, prosody	Voice description	-	-	2022.11
InstructTTS	M	Natural-language style instruction	-	-	-	-	Emotion, style	△	-	-	2023.01
ControlSpeech	M	Reference speech + style prompt	-	-	-	-	Style, timbre	✓	-	-	2024.06
Qwen3-TTS	M	Voice sample / text description	-	-	-	-	Fine-grained style	✓	✓	-	2026.01
PerceptiveAgent	C	User speech + dialogue context	✓	△	-	-	Empathic style	△	△	△	2024.06
Style-Talker	C	Audio + history + style	✓	△	-	-	Speaking style	△	✓	△	2024.08
JELLY	C	Speech emotion + dialogue history	✓	✓	-	-	Emotion	-	-	△	2025.01
RADKA-CSS	C	Current + stored dialogues	✓	△	✓	-	Style, empathy	-	-	△	2025.01
Chain-Talker	C	Dialogue history	✓	✓	-	-	Emotion, empathy	-	-	△	2025.05
Self-EmoQ	C	Dialogue context	✓	✓	-	✓	Planned emotion	-	✓	△	2026.06
Sympatheia	C	User affect / VA signal	✓	△	-	-	Continuous affect	-	△	△	2026.06
AudioGPT	A	Natural-language audio task	△	✓	✓	-	△	-	-	△	2023.04
SpeechAgents	A	Role + multimodal interaction	✓	✓	△	△	✓	✓	△	✓	2024.01
WavCraft	A	Natural-language edit task	△	✓	✓	-	△	-	-	△	2024.03
PodAgent	A	Topic + host/guest roles	✓	✓	✓	△	Role, expressivity	✓	-	△	2025.03
DialogueAgents	A	Character pool + script	✓	✓	△	✓	Emotion, paralinguistics	✓	-	△	2025.04
Qingiang Agents	A	Opera brief + role plan	✓	✓	✓	△	✓	✓	-	△	2025.04
Audiobook Agents	A	Book + scene / role plan	✓	✓	✓	△	✓	✓	-	△	2025.05
AI4Reading	A	Book + editorial goals	✓	✓	✓	✓	△	△	-	△	2025.12
STEAMROLLER	A	Disfluent user speech	✓	✓	△	✓	-	-	✓	✓	2026.01
ActorMind	A	Role + scene + spoken context	✓	✓	-	△	Emotion, verbal traits	✓	-	△	2026.04
AgentSteerTTS	A	Composite instruction	△	✓	✓	✓	Emotion, intensity	✓	-	△	2026.05
AuDirector	A	Narrative + human feedback	✓	✓	✓	✓	Emotion, voice casting	✓	-	△	2026.05
Audio-Oscar	A	Complex scene description	✓	✓	✓	✓	Voice, scene timing	✓	-	△	2026.06
PRISM	A	Prosody + dialogue context	✓	✓	✓	△	Empathic prosody	△	△	✓	2026.06
Harness TTS	A	Explicit / implicit / conflict context	✓	✓	✓	△	Prompt-tool style	✓	✓	△	2026.07
ChipChat	V	Streaming spoken dialogue	✓	△	-	-	-	-	✓	✓	2025.09
Telesales Voice Agent	V	Call audio + playbook	✓	✓	△	△	△	✓	✓	✓	2025.09
SpeechAgent	V	Impaired speech + user needs	✓	✓	△	△	-	△	✓	✓	2025.10
F-Actor	V	Instruction + live context	✓	△	-	-	Voice, behavior	✓	✓	✓	2026.01
LTS-VoiceAgent	V	Streaming user speech + task	✓	✓	△	△	Timing	-	✓	✓	2026.01
PersonaPlex	V	Role prompt + voice sample	✓	△	-	-	Role, style	✓	✓	✓	2026.02
DuplexSLA	V	Audio + action channel	✓	✓	✓	△	Timing, backchannel	△	✓	✓	2026.05
ModeratorLM	V	Role + multi-party context	✓	✓	-	-	Turn behavior	✓	✓	✓	2026.06
VoxMind	V	Spoken task + dialogue state	✓	✓	✓	△	Contextual speech	△	✓	✓	2026.07
JoyAI-Talker	V	Non-verbal cues + persona	✓	✓	△	△	Emotion, rate, volume	✓	✓	✓	2026.08

Interpretation. The decisive shift is not simply from cascaded to end-to-end synthesis. It is the movement of control ownership from the user-supplied condition to a context-sensitive planner, and finally to an online agent that couples speech with timing, tools and actions.

10. Development Trajectory: From Controllable TTS Models to Voice Agents

The trajectory now separates seven capability phases. The added 2023–2025 rows expose the intermediate step that was previously compressed: LLM tool orchestration and multi-agent content production appeared before closed-loop controllable TTS and real-time voice agents.

Phase	Period	Representative milestones	Control shift	Agent mechanism	Open boundary
Latent attribute control	2018–2019	GST-Tacotron; Mellotron	Acoustic/style factors are exposed as controllable latent variables.	No autonomous decision; the user or reference audio supplies control.	Control remains utterance-local and only indirectly interpretable.
Natural-language control	2022–2024	PromptTTS; InstructTTS; PromptTTS 2; ControlSpeech	Text descriptions replace specialist acoustic knobs and scale voice/style prompting.	LLMs encode prompts, but the model still executes a supplied condition.	The system does not decide the appropriate style for the situation.
Tool-orchestrated audio agents	2023–2024	AudioGPT; SpeechAgents; WavCraft	Control moves from one model to an LLM-planned chain of speech/audio tools and roles.	Task planning, model selection and multi-agent communication enter the pipeline.	Speech control is broad and modular, but feedback and real-time interaction are limited.
Context-aware synthesis	2024–2026	PerceptiveAgent; Style-Talker; JELLY; Chain-Talker; Borderless	User audio, history, affect and scene context become control observations.	Inference and retrieval choose how the response or long-form scene should sound.	Persistent planning, external action and reliable self-correction remain partial.
Multi-agent speech production	2025–2026	PodAgent; DialogueAgents; Audiobook Agents; AI4Reading; ActorMind; AgentSteerTTS	Agents plan scripts, roles, voices, timing and expressive attributes across long outputs.	Planner/retriever/critic and proofreader loops make generation auditable and revisable.	Many systems remain offline production pipelines rather than live agents.
Interactive voice agents	2025–2026	Telesales Agent; SpeechAgent; ChipChat; F-Actor; PersonaPlex	Control expands from how to speak to when to speak, how to repair input and how to follow a role/playbook.	Streaming state, dialogue policy and incremental reasoning enter the speech loop.	Long-horizon memory, tool reliability and expressive control remain uneven.
Integrated spoken agents	2026.05–08	DuplexSLA; PRISM; VoxMind; JoyAI-Talker; Harness TTS	Speech, language, tool/action and expressive control share one online decision process.	Reason–plan–act, tool routing and empathetic expression close the model-to-agent loop.	Safety, controllable recovery, persistent memory and realistic evaluation remain open.

Trajectory Synthesis

- **Representation：** latent style tokens → codec/diffusion speech → speech-native and synchronized speech-language-action streams.
- **Control ownership：** explicit user condition → context inference → planner/retriever/critic → online reason-plan-act policy.
- **Temporal coupling：** utterance-level generation → conversational context → streaming/full-duplex speech action.
- **Evaluation：** quality and controllability → context alignment → tool/task correctness → reliability under real-world interaction.

11. Academic Roadmap and Capability Curve

The curve summarizes increasing autonomy, closed-loop control and temporal coupling. Branches expand representative works already listed in this collection; vertical position is conceptual and does not indicate benchmark performance.

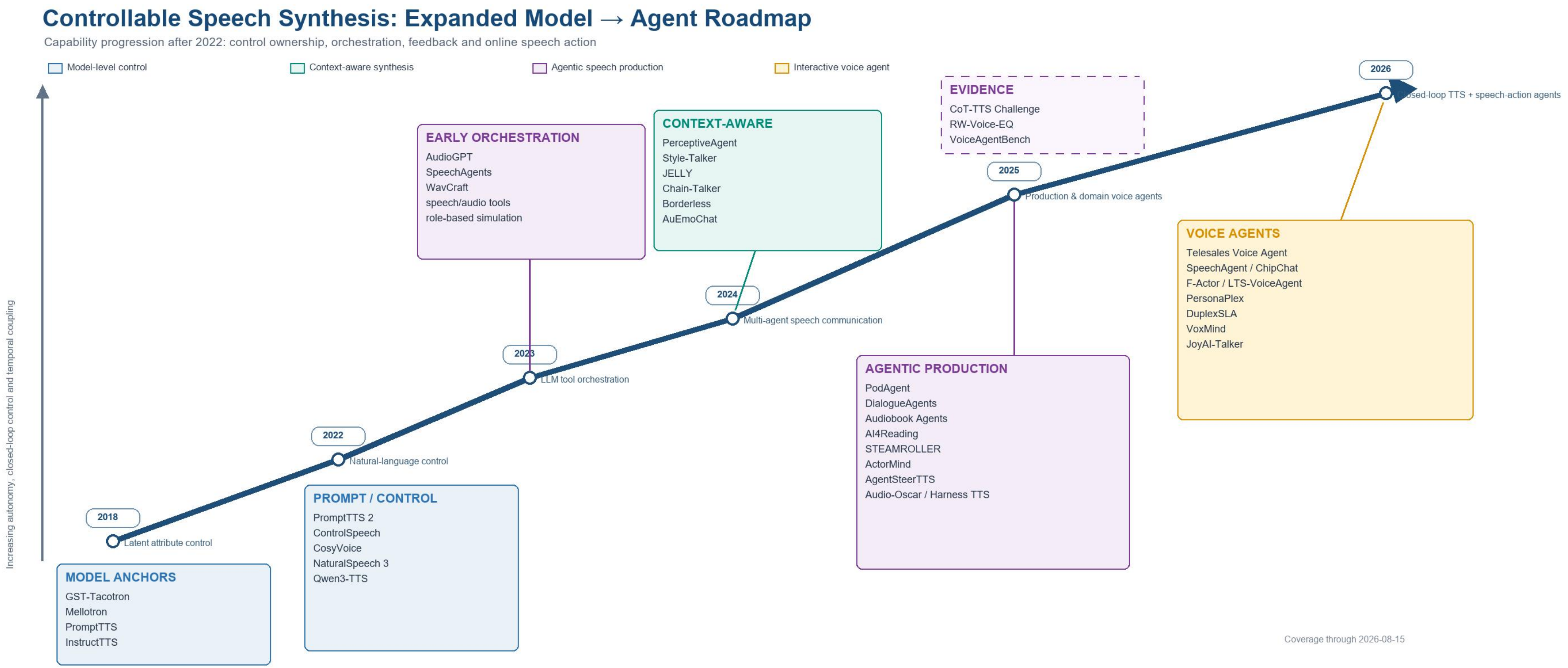

Figure 1. Development roadmap for controllable speech synthesis from model-level control to interactive voice agents. Dashed evidence boxes denote benchmark support rather than model milestones.

Interpretation. The expanded evidence reveals a two-step Agent transition. First, AudioGPT, SpeechAgents and WavCraft (2023–2024) move control from a single acoustic model to an LLM-orchestrated tool or role network. Second, 2025–2026 systems transfer control ownership to planners, retrievers, critics, proofreaders and streaming dialogue policies. The remaining challenge is not simply higher fidelity, but reliable long-horizon control over what to say, how to say it, when to speak and which action should follow.